YUFENG HOU

rainhou12345@gmail.com | 236-967-3159 | Vancouver, Canada

Robotics-focused engineering student at the University of British Columbia with hands-on experience in dexterous robotic hands, robotic arm systems, teleoperation, and simulation-based testing. Skilled in CAD design, Arduino-based prototyping, and data-driven engineering analysis, with a strong record of iterative design, technical documentation, and cross-team collaboration.

EDUCATION

University of British Columbia (UBC) Vancouver, BC
Bachelor of Applied Science in Engineering Expected 06/2029
Relevant Coursework: Engineering Design, Advanced Mathematics, MATLAB Programming, Engineering Physics

TECHNICAL SKILLS

Programming	C, C++, Python, MATLAB
Robotics	Teleoperation, URDF, MuJoCo, ROS, Motion Mapping, Sensor Integration
Mechanical	Fusion 360 CAD, Mechanical Design, Prototyping & Rapid Iteration, Mechanical Testing
Electronics	Arduino, Servos, Breadboard Circuit Development
Other	Data Analysis, Troubleshooting & Testing, Microsoft Excel

ROBOTICS EXPERIENCE

Robotics Engineering & Business Development Intern — Linkerbot *Beijing, China | Summer 2026*

- Supported the development and technical evaluation of multi-DoF dexterous robotic hands and 7-DoF robotic arms.
- Applied robotic teleoperation and human-to-robot motion mapping concepts while supporting joint-control and system-integration work.
- Supported the organization and technical review of URDF-based robotic-hand models and became familiar with MuJoCo simulation workflows.
- Created a voltage-to-joint-angle model from experimental data, including nonlinear curve fitting and uncertainty visualization.
- Produced technical improvement reports covering actuation, mechanical stability, load capacity, lifetime, and thermal performance.
- Supported integration discussions involving ROS, Isaac Sim, MuJoCo, and Franka robotic platforms.
- Communicated technical information through 10+ documents, presentations, and bilingual meeting summaries for international partners.

ENGINEERING PROJECTS

Quadruped Autonomous Robot

- Designed and built a four-legged robotic dog using Arduino (C++), servos, and multiple sensors.
- Programmed coordinated control of eight leg servos and a tail servo to simulate animal-like movement.
- Integrated obstacle- and edge-detection sensors to enable autonomous navigation.
- Improved gait stability, synchronization, and reliability through repeated testing and redesign.

Autonomous Retrieval System

- Collaborated with a multidisciplinary engineering team to design and construct an autonomous claw mechanism.
- Improved grasping reliability and object-collection performance through multiple design iterations.
- Applied engineering design principles, prototyping, testing, and performance evaluation methodologies.
- Achieved a Top 3 finish in the final design competition.

Accessible Seed-Planting Device

- Designed an assistive seed planter for wheelchair users and individuals with limited shoulder mobility.
- Created CAD models in Fusion 360 and contributed to iterative design improvements based on client requirements.
- Applied human-centered design principles and accessibility-focused engineering practices.
- Presented design concepts and findings through a professional engineering poster.

ADDITIONAL EXPERIENCE

Music Producer *3+ years*

- Produced and refined original instrumental compositions using FL Studio, developing audio engineering, digital production, and project management skills.

Graphic Design Volunteer & Content Creator — Youth Development Leadership (YDL)

- Created promotional graphics and edited video content for outreach campaigns, and independently produced and edited YouTube videos end to end.